

and $\llbracket P \rrbracket$ is the set of states that satisfy P in:

$$\llbracket v \rrbracket(s) = s(v)$$

$$\llbracket \text{ff} \rrbracket \triangleq \emptyset$$

$$\llbracket \text{tt} \rrbracket \triangleq \text{State}$$

$$\llbracket P \vee Q \rrbracket \triangleq \llbracket P \rrbracket \cup \llbracket Q \rrbracket$$

$$\llbracket P \wedge Q \rrbracket \triangleq \llbracket P \rrbracket \cap \llbracket Q \rrbracket$$

$$\llbracket P \rightarrow Q \rrbracket \triangleq \llbracket \neg P \vee Q \rrbracket$$

$$\llbracket t_1 = t_2 \rrbracket \triangleq \{s \mid \llbracket t_1 \rrbracket(s) = \llbracket t_2 \rrbracket(s)\}$$

$$\llbracket \forall v. P \rrbracket \triangleq \{s \mid \forall n \in \mathbb{N}. s[v \mapsto b] \in \llbracket P \rrbracket\}$$

$$\llbracket \exists v. P \rrbracket \triangleq \{s \mid \exists n \in \mathbb{N}. s[v \mapsto b] \in \llbracket P \rrbracket\}$$

Partial correctness states that for any asserts P, Q and command C , $\models \{P\} C \{Q\} \triangleq \forall s, s'. [s \in \llbracket P \rrbracket \wedge \langle C, s \rangle \Downarrow s' \Rightarrow s \in \llbracket Q \rrbracket]$.

We can define rules of Hoare logic:

$$\text{H-ASGN} \frac{}{\vdash \{P[E/X]\} \quad x ::= E \quad \{P\}}$$

$$\text{H-SEQ} \frac{\vdash \{P\} C_1 \{Q\} \quad \vdash \{Q\} C_2 \{R\}}{\vdash \{P\} \quad C_1; C_2 \quad \{R\}}$$

$$\text{H-IF} \frac{\vdash \{P \wedge B\} C_1 \{Q\} \quad \vdash \{P \wedge \neg B\} C_2 \{Q\}}{\vdash \{P\} \quad B ? C_1 : C_2 \quad \{Q\}}$$

$$\text{H-WHL} \frac{\vdash \{P \wedge B\} \quad C \quad \{P\}}{\vdash \{P\} \quad \Downarrow B : C \quad \{P \wedge \neg B\}}$$

$$\text{H-CNSQ} \frac{\vdash_L P + s \Rightarrow P_W \quad \vdash_L Q_S \Rightarrow Q_W \quad \vdash \{P_W\} \quad C \quad \{Q_S\}}{\vdash \{P_S\} \quad C \quad \{Q_W\}}$$

Soundness means any triple that can be derived holds semantically:

$$\vdash \{P\} C \{Q\} \Rightarrow \models \{P\} C \{Q\}.$$

Completeness means any triple that holds semantically can be derived:

$$\models \{P\} C \{Q\} \Rightarrow \vdash \{P\} C \{Q\}.$$